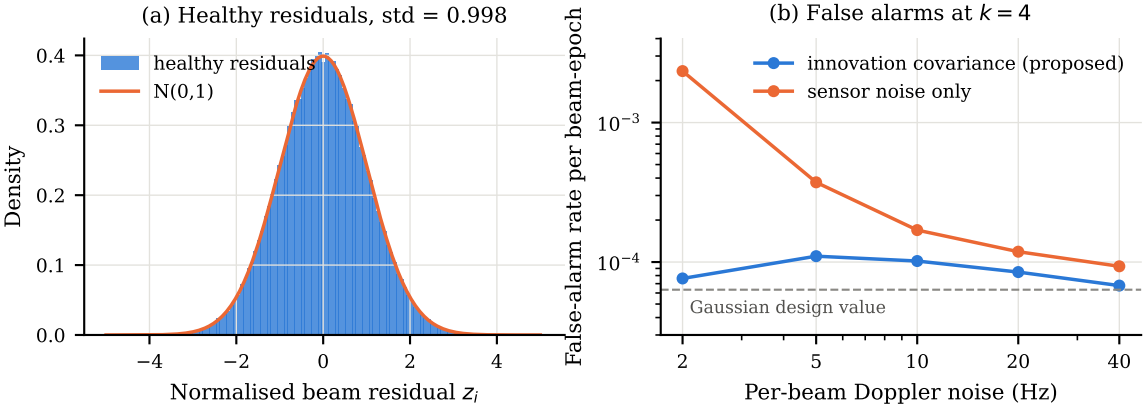
(a) Healthy residuals, std = 0.998
healthy residuals
N(0,1)
Density
0.0
0.1
0.2
0.3
0.4
−4
−2
0
2
4
Normalised beam residual z_i
(b) False alarms at $k = 4$
innovation covariance (proposed)
sensor noise only
10^{-3}
10^{-4}
Gaussian design value
False-alarm rate per beam-epoch
2
5
10
20
40
Per-beam Doppler noise (Hz)